Department of Statistics & Computer Science
University of Kelaniya
ACADEMIC YEAR – 2022/2023
COSC 22083/BECS 22243 -Database Management Systems

Assignment 03

01.

A. Write short answers for each of the following:

a. What are the components of a disk block address?

Cylinder Number
Track Number
Block Number

b. List the information that is included in the file header.

Disk Address
Format Description

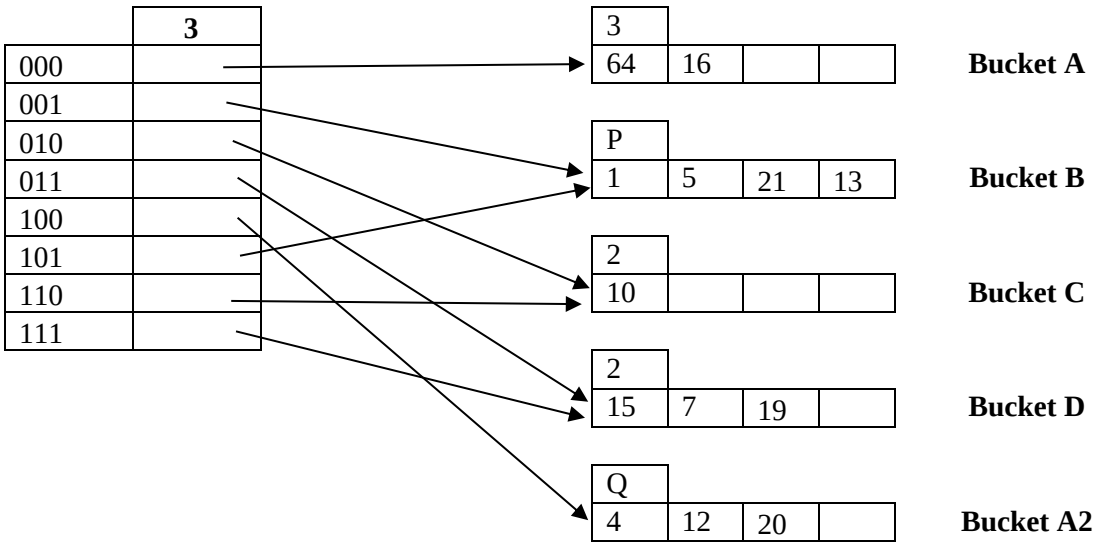
c. How is a hashed file organized?

Based on hashing techniques

d. What is the purpose of linear hashing?

No need a directory

B. Consider the following extendible hashing index diagram. You may assume that the entries in the hash values.



Using the index answer the following questions:

a. **What is the value of the global depth?**

   Global Depth = 3

b. **What are the values of P & Q?**

   P = 2 and Q = 3

c. **What is the use of the local depth?**

   To determine whether a directory doubling is needed we maintain a local depth for each bucket. (To overflow handling)

d. **Is it possible to give that last entry that was inserted into the index? Give reasons.**

   No. There is not given the inserting order.

e. **Is it possible to say that a split has occurred due to an insertion of any entry? If so, what could be the possible entry that was inserted.**

   Yes

   64, 16, 4, 20, 12

f. **List the steps to search the hash value 20.**

   Binary code of 20 is – 10100

   Check the global depth. It is 3 so check last 3 digits. Corresponding address is bucket A2. So search A2

g. **By using a diagram show the index after inserting the entry with hash value 9. Justify your answer.**

   Binary code of 9 is – 1001

   This must be inserting inti bucket B. But it is full.

   Local depth = 2 and Global depth =3

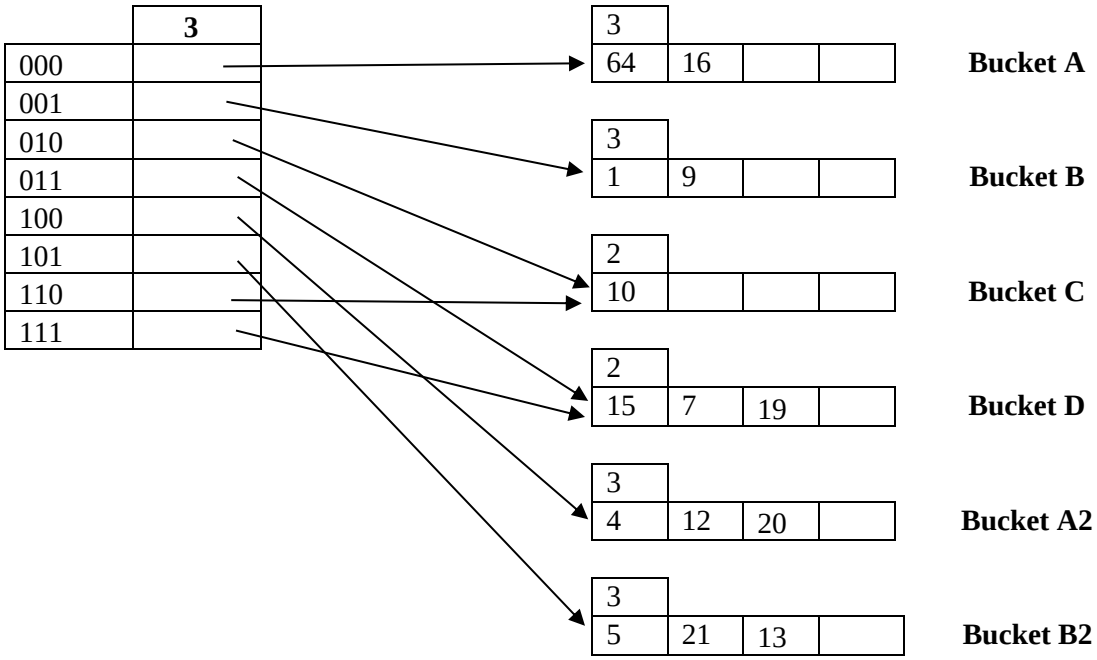
   Local depth != Global depth. So not double the directory

   Split the bucket B.

   Then restore the entries in the bucket B within bucket B and bucket B2 checking their binary codes.

Number	**Binary Code**	**Bucket**
1	**001**	Bucket B
5	**101**	Bucket B2
21	10**101**	Bucket B2
13	1**101**	Bucket B2

Following index show after inserting the entry with hash value 9.



02.

A. List two differences between 'Extendible Hashing' and 'Linear Hashing'.

Extendible Hashing	**Linear Hashing**
Uses a type of directory	No need a directory
Define local depth and global depth	Define level and next pointer

B. **Consider the following linear hashing index. Using this index, answer the following questions.**

a. Briefly describe the given linear hashing index.

This is a level = 0 and number of buckets initially = 4. But there are 5 buckets and their split has occurred.

There is a hash function family h0 and h1 when overflow page is added split the bucket which is next pointer is point out.

b. Show the bucket that the Next pointer is pointing at.

Level = 0 N = 4

h1	h0
000	00
001	01
010	10
011	11
100	00

Next = 1

PRIMARY PAGE **OVERFLOW PAGE**

32	8	24	
9	25	41	17
14	18	10	30
31	35	7	11
44	36		

c. Is it possible to give the last entry that was inserted into the index? Give reasons.

No. Because the order of inserting is not given.

d. Is it possible to say that a split has occurred due to an insertion of any entry? If so, what could be the possible entry that was inserted.

Yes. 32, 8, 24, 44, 36.

e. For the above given index, what is the minimum number of data entries that can be inserted before a bucket is split. Explain briefly. You assume that the possible distribution of keys is given.

3. There are spaces in bucket 1 and 5. So we can insert 3 values without splitting the bucket.

f. Show the index after inserting the entries 4 and 15 respectively. Justify your answer.

inserting value 4

Binary code of 4 =100

Consider last 2 digits = 00 (Because level = 0) but bucket has split so consider last 3 digits 100 goes to last bucket. It has space so does not split the bucket. So Next =1.

After inserting value 4

Level = 0 N = 4

PRIMARY PAGE **OVERFLOW PAGE**

h1	h0
000	00
001	01
010	10
011	11
100	00

Next = 1

32	8	24	
9	25	41	17
14	18	10	30
31	35	7	11
44	36	4	

inserting value 15

Binary code of 15 =1111

Consider the last 2 digits = 11 (Because level = 0). So go to bucket 4. But it is full. So, an overflow page is added. Because the overflow page is added so split the bucket which Next is pointing at. So added has family 101 01.

Then Next pointer move to the next bucket.

So, we again check the last 3 digits of bucket 2 number. The binary codes of 9, 25, 41, 17

Number	**Biinary Code**	**Hash Family**
9	1**001**	001
25	11**001**	001
41	101**001**	001
17	10**001**	001

After inserting value 15

Level = 0 N = 4

PRIMARY PAGE **OVERFLOW PAGE**

h1	h0
000	00
001	01
010	10
011	11
100	00
101	01

Next = 2

32	8	24	
9	25	41	17
14	18	10	30
31	35	7	11
44	36	4	

15			

Submission Guidelines:

- Make sure to **attach a cover page** starting with the course code, course name, assignment number, and your student number.
- Please submit **a hard copy** to the Department of Statistics & Computer Science.
- **Deadline for Submission**: On or before **17th of January 2025**.